

* SETS

- intersection $\rightarrow \cap$
- Union difference $\rightarrow \cup$
- Symmetric differences $\rightarrow A \Delta B$
- Universal set

• Complement of set

• Empty (NULL set) $\rightarrow \emptyset$ $S = \{1, 2, 3, 5, 7, 8, 9, 4\} \Rightarrow$ SAMPLE SPACE (UNIVERSAL SET)

$A = \{2, 3, 5, 1\}$, $B = \{1, 3, 7, 8, 9\}$

Find

1- $A \cap B = \{1, 3\}$

4- $B - A = \{7, 8, 9\}$

2- $A \cup B = \{2, 3, 5, 1, 7, 8, 9\}$

5- $A \Delta B = (A - B) \cup (B - A)$

3- $A - B = \{2, 5\}$

$= \{2, 5, 7, 8, 9\}$

الخاصة الموجودة في A ولا توجد في B

5

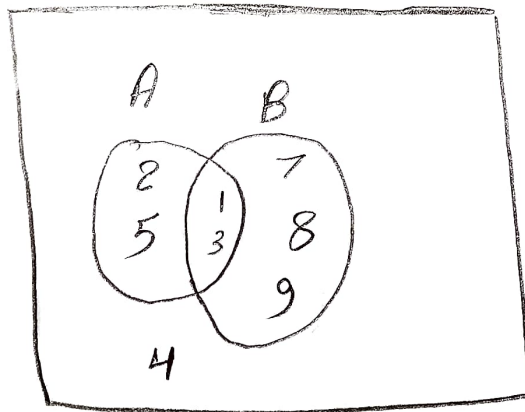
$A \subseteq S$ subset

المجموعة A جزئية من المجموعة S

$\{2, 3, 5, 1\} = A$

$\{3, 7\} \subseteq B$

$9 \in B$



$\rightarrow$ Venn diagram

$\neq$ not equal

$\not\subseteq$ not subset

$\in$ belongs to

$\notin$ doesn't belong to

* empty set $\emptyset$

$$\textcircled{1} (A^c)^c = A$$

$$\textcircled{2} \emptyset^c = S$$

$$\textcircled{3} S^c = \emptyset$$

$$\textcircled{4} (A \cap B)^c = A^c \cup B^c$$

$$\textcircled{5} (A \cup B)^c = A^c \cap B^c$$

$$\textcircled{6} A - B \neq B - A$$

$$\textcircled{7} A \Delta B = B \Delta A$$

AND $\Rightarrow \wedge$

OR $\Rightarrow \vee$

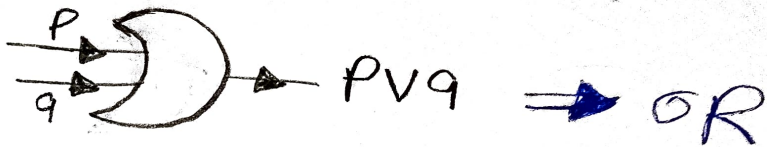
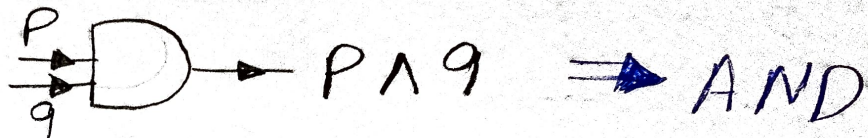
IF ... then $\Rightarrow \rightarrow$

Not $\Rightarrow \sim$

NAND $\Rightarrow \sim(\wedge)$

NOR $\Rightarrow \sim(\vee)$

XOR $\Rightarrow \oplus$



1 $\rightarrow$ True

0 $\rightarrow$ False

equivalent $\Rightarrow \equiv$

* iff = if and only if $\Rightarrow$



P	q	$P \wedge q$	$P \vee q$	$\neg P$	$\neg q$	$P \rightarrow q$	$q \rightarrow q$	$P \leftrightarrow q$	$\neg(P \wedge q)$	$\neg P \vee \neg q$
1	1	1	1	0	0	1	1	1	0	0
1	0	0	1	0	1	0	1	0	1	1
0	1	0	1	1	0	1	0	0	1	1
0	0	0	0	1	1	1	1	1	1	1

P	q	$P \wedge q$	$q \vee r$	$P \wedge (q \vee r)$	$q \wedge r$	$P \vee (q \wedge r)$
1	1	1	1	1	1	1
1	0	0	1	0	0	1
1	1	1	1	1	1	1
0	0	0	0	0	0	0
0	1	0	1	0	0	1
0	0	0	0	0	0	0
0	1	0	1	0	0	1

$$P \wedge (q \vee r) \neq P \vee (q \wedge r)$$

If $x \in A \rightarrow 1$ or True "T"

If $x \notin A \rightarrow 0$ or False "F"

A	True "agree"
1	False "disagree"
0	

For two sets A, B

① $A \cap B$

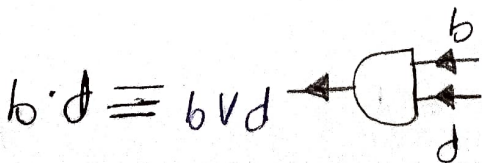
② $A \cup B$

③ $A - B$

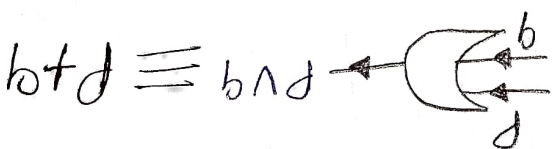
A	B	$A \cup B$	$A \cap B$	$A - B$	A'	B'
1	1	1	1	0	0	0
1	0	1	0	1	0	1
0	1	1	0	0	1	0
0	0	0	0	0	1	1

* Logic gates *

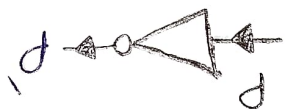
AND gate



OR gate



AND $\equiv \odot$
OR $\equiv \oplus$



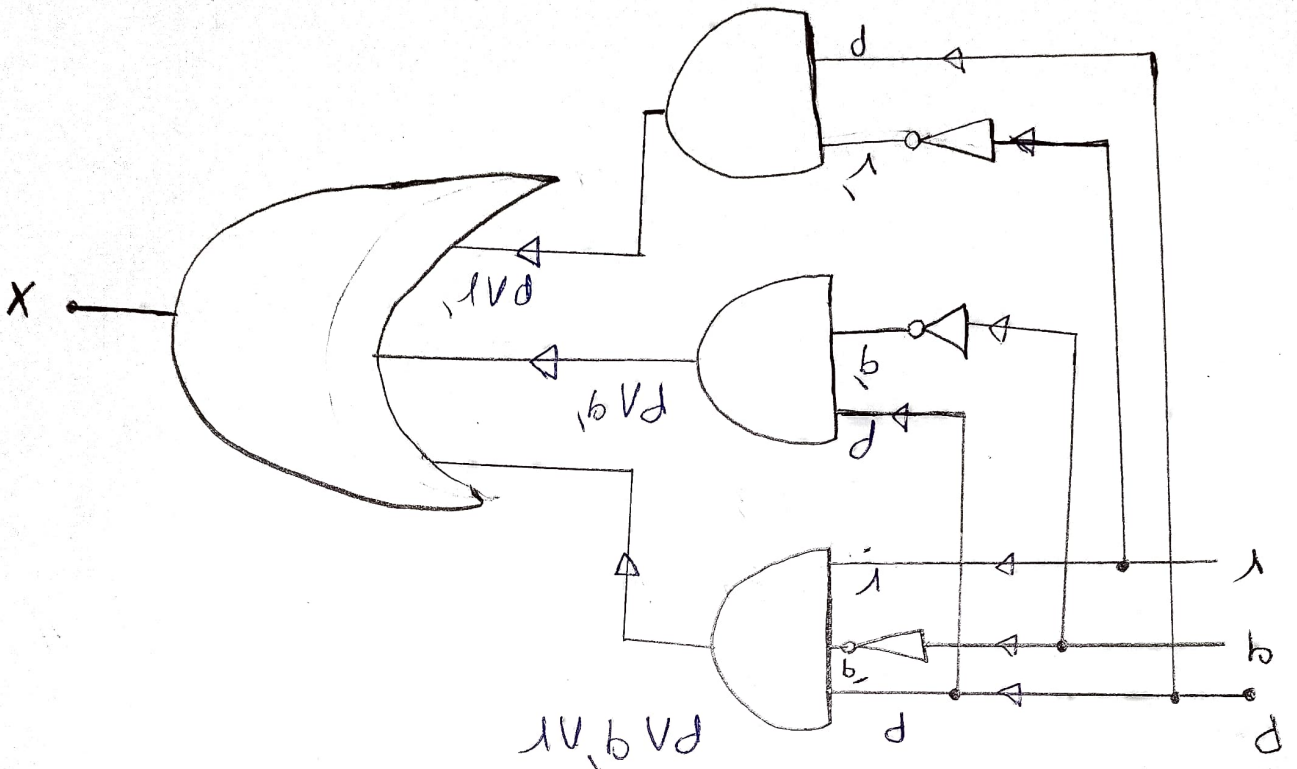
Not gate

* $p' = \sim p$



NOR gate

Ex: write the output X.



 $\therefore X \equiv (p'q'r) \vee (p'q'r') \vee (p'r')$

*BASIC RULES OF BOOLEAN ALGEBRA

$$A + 0 = A$$

$$A + 1 = 1$$

$$A \cdot 0 = 0$$

$$A \cdot 1 = A$$

$$A + A = A$$

$$A + \bar{A} = 1$$

$$A \cdot A = A$$

$$A \cdot \bar{A} = 0$$

$$\bar{\bar{A}} = A$$

$$A + AB = A$$

$$A + \bar{A}B = A + B$$

$$(A+B)(A+C) = A + BC$$

De Morgan's Theorem

$$(\overline{AB}) = (\bar{A} + \bar{B})$$

$$\overline{(A+B)} = (\bar{A} \bar{B})$$

~~EX: F = AB + AB~~

$$\begin{aligned} \text{EX: } F &= AB + AB \\ &= AB(1 + 1) = AB \end{aligned}$$

$$\rightarrow \bar{V} = \overline{A + BC}$$

$$= \bar{A} \cdot \bar{BC}$$

$$\bar{V} = \bar{A} \cdot (\bar{B} + \bar{C})$$

$$F = A(\bar{A} + B)$$

$$F = \cancel{A\bar{A}} + AB$$

$$F = AB$$

$$F = \cancel{AA} + AC + \cancel{AB} + BC$$

$$= A + AC + AB + BC$$

$$= A(\underbrace{1 + C + B}_1) + BC$$

$$F = A + BC$$

#

$$F = A + AB$$

$$= A(1 + B)$$

$$F = A$$

$$F = AB + A\bar{B}$$

$$= A(\bar{B} + B)$$

$$F = A$$

$$F = AB + BC(B + C)$$

$$F = AB + \cancel{B\bar{B}C} + \cancel{B\bar{B}C}$$

$$= AB + BC + \cancel{BC}$$

$$F = AB + BC$$

$$F = B(A + C)$$

$$F = AB + A(A + C) + B(A + C)$$

$$F = AB + \cancel{AA} + AC + \cancel{BA} + BC$$

$$= AB + A + AC + BC$$

$$= A(\underbrace{B + 1 + C}_1) + BC$$

$$F = A + BC$$

#

$\downarrow$ A	B	C	$B \vee C$	$\downarrow$ $\neg(B \vee C)$	$A \wedge \neg(B \vee C)$
1	1	1	1	0	0
1	1	0	1	0	0
1	0	1	1	0	0
1	0	0	0	1	1
0	1	1	1	0	0
0	1	0	1	0	0
0	0	1	1	0	0
0	0	0	0	1	0

A	B	C	$B \wedge C$	$A \wedge C$	$A \wedge B$	$A \vee (B \wedge C)$	$(A \vee B) \wedge (A \vee C)$
1	1	1	1	1	1	1	1
1	1	0	0	1	1	1	1
1	0	1	0	1	1	1	1
1	0	0	0	1	1	1	1
0	1	1	1	1	1	1	1
0	1	0	0	0	1	1	1
0	0	1	0	1	0	0	0
0	0	0	0	0	0	0	0

$$\therefore A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$$

Kernough maps

$$\rightarrow P \vee q$$

$$\Downarrow$$

$$\underline{P+q}$$

$$P \wedge q$$

$$\Downarrow$$

$$\underline{P \cdot q}$$

$$= \rightarrow =$$

* Express on this logical statement by boolean Algebra

$$X = (P \wedge q) \vee (P \wedge q')$$

$$X = P \cdot q + P \cdot q' \neq$$

$$= P (q + q') = P$$

*Kernough maps for 2 switch

	q	q'
P	Pq	Pq'
P'	P'q	P'q'

* reduce this fig. using kernough maps

	q	q'
P	(X)	
P'	(X)	(X)

$$X = Pq + P'q + P'q'$$

Solution

* كل رقم موجود فقط في المربع بناءً على (X)

$$X = q + p'$$

* K. maps for 3 switch P, q, r

	qr	q'r	q'r'	q'r
P	Pqr	Pq'r	Pq'r'	Pq'r
P'	P'qr	P'q'r	P'q'r'	P'q'r

Ex: reduce the circler represented by this expression

$$X \equiv (p \wedge q \wedge r) \vee (p' \wedge q \wedge r') \vee (p \wedge q' \wedge r)$$

$$X = pqr + p'qr' + pq'r$$

	qr	q'r'	q'r'	q'r
p	X			X
p'		X		

$$X = pr + p'qr'$$

Ex: $X = pqr + pqr' + p'qr' + pq'r$

* ازاى

جاء

∴

* لوعا

ال

∴

	qr	qr'	q'r'	q'r
p	X	X		X
p'		X		

$$X = pq + qr' + pqr'$$

$$X = qr' + pr$$

⊆

	qr	qr'	q'r'	q'r
p	X	X		X
p'		X		

TYPE OF RELATIONS

① REFLEXIVE RELATION علاقة

R is said to be reflexive on A

$$\forall x \in A \quad x R x$$

EX : IF $A = \{a, b, c, d\}$

$$R_1 = \{(a, a), (b, b), (c, d), (c, c), (d, d), (a, d)\}$$

Since $a R a, b R b, c R c, d R d$

therefor R_1 is reflexive relation

② IRREFLEXIVE RELATION غير عاكسة

$$R_2 = \{(a, b), (b, b), (d, d), (d, b)\}$$

$\because c \notin R_c \therefore R_2$ is irreflexive relation

ليس له علاقة
بنفسه
if $\exists$

there exist an element or more from A is not related itself

③ Symmetric Relation علاقات متماثل

R is said to be symmetric relation on A

$$\forall a, b \in A \quad a R b \Rightarrow b R a$$

Ex IF $A = \{a, b, c, d\}$

$$R_3 = \{(a, b), (c, d), (d, b), (c, c), (b, b), (a, a)\}$$

→ R_3 is not reflexive.

→ R_3 is symmetric.

$$R_4 = \{(a, b), (c, d), (d, d), (b, a)\}$$

→ R_4 is not reflexive

→ R_4 is not symmetric

④ Transitive Relation

$$\forall a, b, c \in A$$

$$a R b, b R c \Rightarrow a R c$$

$$R_5 = \{(c, d), (b, b), (d, e), (c, e), (e, e)\}$$

→ R_5 is not reflexive

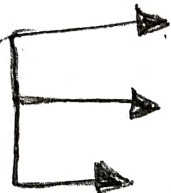
→ R_5 is symmetric

→ R_5 is transitive because $c R d, d R e \Rightarrow c R e$

⑤ Equivalence Relation

علاقة تكافؤ

if its



Symmetric

Reflexive

Transitive



① ~~is~~

= on N

$n = 1, 2, 3, \dots$

②

< on N

③

> on N

④

// Parallel

Solution

① ~~is~~ =

is Reflexive ($1=1, 2=2, \dots$)

~~is~~ =

is not symmetric

EX: If $A = \{e, b, c, d, e, f\}$

$R = \{(e, e), (e, b), (e, c), (b, e), (b, b), (b, c), (c, e), (c, b), (c, c), (d, d), (d, e), (e, d), (e, e), (f, f)\}$

① ~~IS~~ IS R is equivalence Relation

② write M_R

③

Solution

- ①
- R is REFLEXIVE
 - R is SYMMETRIC
 - R is TRANSITIVE

* R is equivalence

* إذا كانت عناصر القطر الرئيسى

جميعها 1

∴ العلاقة REFLEXIVE

* لو كان جانبا القطر

الرئيسى = 1

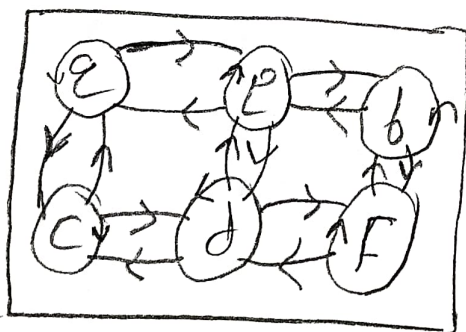
∴ SYMMETRIC

② $M_R =$

	e	b	c	d	e	f
e	1	1	1	0	0	0
b	1	1	1	0	0	0
c	1	1	1	0	0	0
d	0	0	0	1	0	0
e	0	0	0	0	1	0
f	0	0	0	0	0	1

③

Arrow diagram



Let R_1 and R_2 are defined on $A = \{1, 2, 3\}$

$$R_1 = \{(1,1), (1,2), (2,3), (3,1), (3,3)\}$$

$$R_2 = \{(1,2), (2,1), (2,3), (3,1)\}$$

$$R_1 \cup R_2 = \{(1,1), (1,2), (2,3), (3,1), (3,3), (2,1)\}$$

$$R_1 \cap R_2 = \{(1,2), (2,3), (3,1)\}$$

$$R_1 - R_2 = \{(1,1), \text{[scribble]}, (3,3)\}$$

$$R_2 - R_1 = \{(2,1)\}$$

$$R_1 \oplus R_2 = \{(1,1), (3,3), (2,1)\}$$

$$R_1^{-1} = \{(1,3), (2,1), (2,2), (3,2)\}$$

$$R_2^{-1} = \{(1,1), (1,3), (2,2), (3,2), (3,3)\}$$

$$R_1^{-1} = \{(1,1), (2,1), (3,2), (1,3), (3,3), (1,2)\}$$

$$R_2^{-1} = \{(2,1), (1,2), (3,2), (1,3)\}$$

$$R_2 \circ R_1 = \{(1,2), (1,1), (1,3), (2,1), (3,2), (3,1)\}$$

$$R_1 \circ R_2 = \{(1,3), (2,1), (2,2), \text{[scribble]}, (2,3), (3,1), (3,2)\}$$

$$R^{\circ} = \{(1,1), (2,2), (3,3)\}$$

$$R_1^2 = R_1 \circ R_1 = \{(1,1), (1,2), (1,3), (2,1), (2,3), (3,1), (3,2), (3,3)\}$$

R and S be two relation over the set A and B,
respectively

• Intersection of Relation:

$$M_{R \cap S} = M_R \odot M_S \Rightarrow$$

خرب العناصر المتناظرة في
المصفوفتين وهي عليه
تختلف عن ضرب المصفوفات العادية

• Union

$$M_{R \cup S} = M_R \oplus M_S$$

Note:

$$\begin{aligned} \rightarrow 0 + 0 &= 0 \\ \rightarrow 1 + 0 &= 0 + 1 = 1 \\ \rightarrow 1 + 1 &= 1 \end{aligned}$$

• Difference

$$R - S \neq S - R$$

• العناصر الموجودة في S ولا توجد
في R والعكس صحيح

• Symmetric Difference

$$\rightarrow R \Delta S = (R \cup S) - (R \cap S)$$

$$\rightarrow R \Delta S = (R - S) \cup (S - R)$$

• إذا كانت R علاقته انعكاسية فـ R^{-1} متماثلة أيضاً
• " " R " " متماثلة " " متماثلة " "
• " " R " " تكافؤ " " تكافؤ " "
• اتحادا وتقاطع علاقيتين متماثلتين ينتج عنه
علاقته متماثلة أيضاً.

• Complement

$$\bar{R} = R^c = \hat{R}$$

هي العلاقة التي تحتوي على جميع
الازواج الممكنة التي لا تنتمي الى R

• ~~converse~~ / INVERSE

يتم الحصول على العلاقة العكسية R^{-1} / R^T عن طريق تبديل
ترتيب العناصر في كل زوج مرتب

• إذا كانت العلاقة ممثلة بمصفوفة M_R فيكون مصفوفة العلاقة العكسية
عن مدور المصفوفة M_{R^T} (تبديل الصفوف بالاعمدة)

Ex: Suppose we have ~~set~~ sets:

$A = \{0, 1, 2\}$ and the relation R and S defined

as follows: Relation $R: \{(0, 2), (1, 0), (1, 2), (2, 0)\}$

Relation $S: \{(1, 0), (1, 1), (1, 2), (2, 2)\}$

$$\rightarrow R \cup S = \{(0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 2)\}$$

$$M_{R \cup S} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \oplus \begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \checkmark$$

$$\rightarrow R \cap S = \{(1, 0), (1, 2)\}$$

$$M_{R \cap S} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \otimes \begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow R^{-1} / R^T$$

$$R_R^T = \{(2, 0), (0, 1), (2, 1), (0, 2)\}$$

$$R_S^T = \{(0, 1), (1, 1), (2, 1), (2, 2)\}$$

$$\rightarrow R_R^{-1} = \{(0, 0), (0, 1), (1, 1), (2, 1), (2, 2)\}$$

$$R_S^{-1} = \{(0, 1), (2, 1), (0, 2), (2, 0), (0, 0)\}$$

Suppose $A = \{a, b, c\}$ and $B = \{1, 2, 3\}$ relations R and S have the form

$$R = \{(a, 1), (b, 2), (c, 3), (d, 1)\} \quad S = \{(a, 1), (b, 1), (c, 1), (d, 1)\}$$

then the relation differences $R - S$ and $S - R$ are given by and find $R \Delta S$

$$R - S = \{(b, 2), (c, 3)\}$$

$$S - R = \{(b, 1), (c, 1)\} \checkmark$$

$$R \Delta S = \{(b, 2), (c, 3), (b, 1), (c, 1)\}$$

Ex: $R = \{(1,2), (1,3), (1,4)\}$ then the converse of R is
 $R^T = \{(2,1), (3,1), (4,1)\}$

Ex: $M_R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \Rightarrow M_{R^T} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$